

Rungta College of Engineering & Technology. Bhilai
Department of Computer Science & Engineering

Project Synopsis

Semester: 6th

1. Project Title: Automated Threat Detection Using Log Correlation in Simulated Enterprise Attacks

2. Project Type: Cybersecurity

3. Project Overview/ Abstract:

The rapid growth of enterprise networks and cloud-based infrastructures has increased the frequency and sophistication of cyberattacks. Organizations generate massive volumes of security logs from firewalls, servers, operating systems, and network devices, making manual monitoring and analysis difficult. This project, titled “**Automated Threat Detection Using Log Correlation in Simulated Enterprise Attacks,**” aims to design and implement a centralized threat detection framework capable of collecting, correlating, and analyzing security logs in real time.

The system utilizes the ELK Stack (Elasticsearch, Logstash, and Kibana) along with Filebeat agents to aggregate logs from multiple systems and detect suspicious activities through log correlation techniques. Simulated enterprise attacks such as brute-force login attempts, unauthorized access attempts, and malicious network activities are generated in a controlled environment to evaluate the effectiveness of the system.

The project focuses on identifying attack patterns, visualizing security events through dashboards, and improving incident response time by automating threat detection. The framework provides organizations with a scalable and efficient method for monitoring enterprise security events and detecting potential threats proactively.

4. Problem Statement:

Modern organizations face significant challenges in identifying and responding to cyber threats due to the large amount of log data generated across enterprise systems. Traditional monitoring methods rely heavily on manual analysis, which is time-consuming, inefficient, and prone to human error.

Existing systems often fail to correlate logs from multiple sources effectively, resulting in delayed detection of suspicious activities and increased security risks. Attackers exploit these gaps to perform brute-force attacks, privilege escalation, unauthorized access, and lateral movement within networks.

There is a need for an automated, centralized, and scalable threat detection system capable of collecting logs from multiple systems, correlating events, and identifying malicious activities in real time.

This project addresses these challenges by implementing an automated threat detection framework using log correlation techniques and enterprise attack simulations.

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5. Objective:

- ☐ To design and implement a centralized log management and threat detection system.
- ☐ To collect and analyze logs from multiple enterprise systems using the ELK Stack.
- ☐ To perform real-time log correlation for detecting suspicious activities.
- ☐ To simulate enterprise cyberattacks in a controlled environment.
- ☐ To visualize security events and attack patterns using Kibana dashboards.
- ☐ To improve incident response time through automated threat detection.
- ☐ To provide a scalable and efficient cybersecurity monitoring framework.

6. Methodology:

The development of the project follows a structured cybersecurity implementation methodology:

- **Environment Setup:**

A virtual enterprise environment is created using Ubuntu virtual machines and Docker containers to simulate enterprise infrastructure.

- **Log Collection:**

Filebeat agents are configured on target systems to collect authentication, system, and network logs.

- **Log Processing & Storage:**

Logstash processes incoming logs and forwards them to Elasticsearch for indexing and storage.

- **Threat Detection & Correlation:**

Log correlation rules and filters are implemented to identify suspicious patterns such as repeated failed login attempts, unauthorized access, and abnormal activities.

- **Visualization:**

Kibana dashboards are used to monitor events, visualize attack patterns, and analyze security incidents in real time.

- **Attack Simulation:**

Controlled enterprise attack scenarios are executed to test the effectiveness of the detection framework.

- **Evaluation:**

System performance is analyzed based on detection accuracy, log visibility, alert generation, and monitoring efficiency.

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7. Flow Chart/DFD/ER Diagram:

- Network Diagram:



Fig: Network Diagram

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- Flowchart

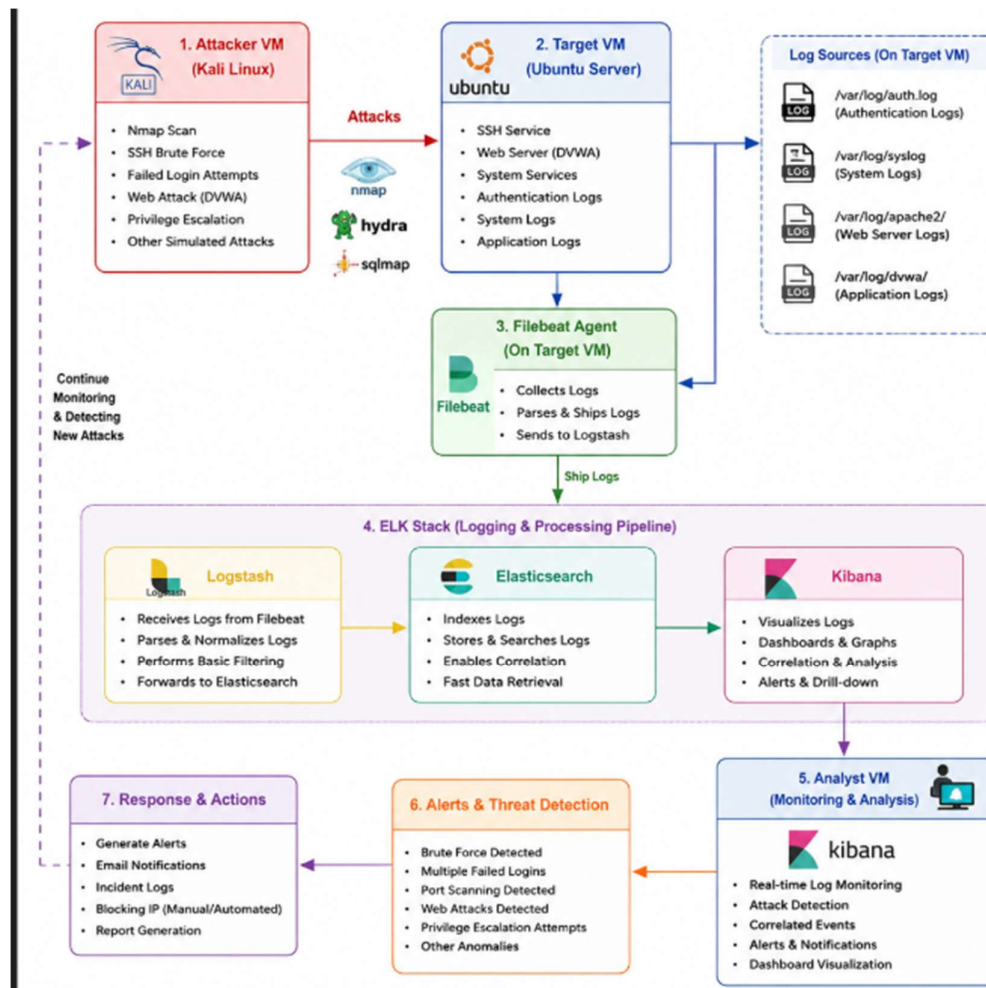


Fig2. Flowchart

8. Technical Details:

- Platform Type:** Web-based Security Monitoring & Threat Detection System

TECHNOLOGIES USED

- ☐ Ubuntu Linux
- ☐ Docker
- ☐ Elasticsearch
- ☐ Logstash
- ☐ Kibana
- ☐ Filebeat
- ☐ Linux Authentication Logs
- ☐ Bash Commands & Network Utilities

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9. Team Members:

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